MTH302:PROBABILITY AND STATISTICS

L:3 T:1 P:0 Credits:4

Course Outcomes: Through this course students should be able to

CO1 :: recall the concept of random variables and its distribution functions.

CO2 :: recognize relationships among the variables through correlation and regression.

CO3 :: apply probability distributions to find the solution of different engineering problems.

CO4 :: analyze estimators for unbiasedness, consistency, and efficiency, including maximum likelihood estimators

CO5 :: interpret the results of statistical tests such as Z-test, t-test, F-test, and Chi-square test for hypothesis testing

Unit I

Random Variables and Probability Distributions : discrete and continuous random variables and their probability distributions, cumulative distribution functions, moments about origin and mean (central moments) up to fourth order, mean, variance, skewness, kurtosis

Unit II

Correlation and Linear regression : scatter plots, correlation coefficient and its properties, Karl Pearson's correlation coefficient, Spearman's rank correlation coefficient, Linear regression and its properties

Unit III

Special Discrete Distributions : bernoulli process, binomial distribution, negative binomial distribution, geometric distribution, poisson distribution, moment generating function (without proof) of the above mentioned distributions.

Unit IV

Special Continuous Distributions : exponential distribution, gamma distribution, normal distribution, normal approximation to the binomial, central limit theorem (without proof), moment generating function (without proof) of the above mentioned distributions.

Unit V

Point Estimation : unbiased estimator, consistent estimator, efficient estimator, maximum likelihood estimation

Unit VI

Testing of Hypothesis : types of error, Z-test for single mean and difference of means, Student's t-test for single mean and difference of means, F-test, Chi-square test for goodness of fit

Text Books:

1. PROBABILITY AND STATISTICS FOR ENGINEERS AND SCIENTISTS by RONALD E. WALPOLE, RAYMOND H. MYERS,SHARON L. MYERS, AND KEYING YE, PEARSON

References:

1. PROBABILITY AND STATISTICS FOR ENGINEERS by DR. J. RAVICHANDRAN, WILEY

2. FUNDAMENTALS OF MATHEMATICAL STATISTICS by S.C. GUPTA AND V.K. KAPOOR, SULTAN CHAND & SONS (P) LTD.